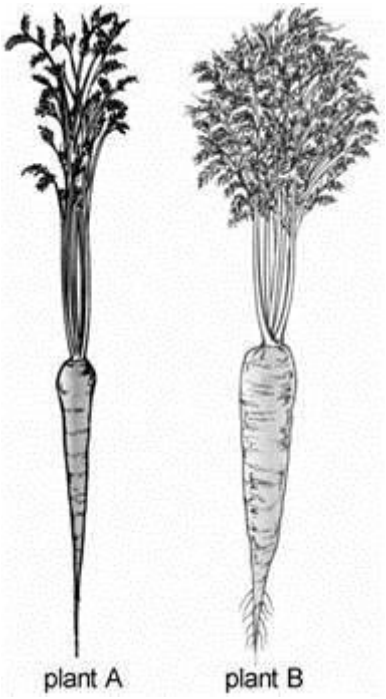
Q1. Some pupils grew carrot plants for a project on plant growth. At the end of the summer they dug up the carrots. The drawings show two of their carrots.



(a) Plant A came from a part of the garden which was covered with weeds. Plant B came from a part of the garden which had been kept free of weeds.

Suggest two ways in which the weeds may have stopped plant A from growing as large and healthy as plant B.

1. .....

.....

2. .....

.....

2 marks

(b) Explain why the pupils' plants produced bigger roots when they received more light.

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3 marks
Maximum 5 marks

Q1.

(a) **answers should imply competition for resources between weeds and carrot plant A**

any **two** from

- the weeds shaded the carrot plants
  accept 'the weeds stopped the Sun getting to the carrots'

- the weeds left the carrots with less space to grow
  *accept 'weeds choked the carrots' **or** 'weeds took up all the space' **or** 'carrot plant A had less space'*
  *do **not** accept 'the carrot plants had less space'*

- the weeds absorbed **or** took most of the available water
  accept 'the weeds took the water'
  ***or** 'carrot plant A had less water'*

- the weeds absorbed **or** took most of the available mineral salts **or** nutrients
  accept 'the weeds took the minerals
  ***or** nutrients **or** nitrates' **or** 'carrot plant A had less minerals'*
  *do **not** accept 'weeds took the food'*

2 (L5)

(b) **there are three marking areas: photosynthesis, carbohydrate production, and carbohydrate transfer or storage. At least one of these areas must contain the term 'more' to gain the three marks**

- more photosynthesis took place
  accept 'photosynthesis took place'
  ***or** 'the plants made more food'*

1 (L6)

- more carbohydrate was produced in the leaves
  *accept 'carbohydrate **or** sugar **or** glucose was made in the leaves'*
  *do **not** accept 'food was made in the leaves' for this mark*

1 (L6)

- more carbohydrate was transported to the roots **or** stored in the roots
  *accept 'carbohydrate **or** sugar **or** glucose **or** food was transported to the roots'*
  *accept 'carbohydrate **or** food **or** starch was stored in the roots'*

1 (L6)

[5]